

AMC 10 Crash Course • Block 2 • Geometry

Theory, worked examples, and independent sets. Solutions to the independent sets are at the end of the file.

Lesson 2.1. Angles: Parallel Lines, Triangles, Polygons

THEORY

Parallel lines. A transversal across two parallel lines produces three families of angle pairs: corresponding angles are equal, alternate interior angles are equal, and same-side interior angles add to 180° . On a contest this should work in one mental move: see parallel lines — slide the angle along the transversal instead of recomputing. If a point sits BETWEEN two parallel lines, draw an auxiliary parallel line through it: any bent angle splits into two simple ones.

Angle sum of a triangle = 180° . An exterior angle = the sum of the two remote interior angles. Angle sum of an n -gon = $(n - 2) \cdot 180^\circ$; the exterior angles of any convex polygon sum to 360° .

Triangles. 180° is the main currency. The exterior angle saves a step: it directly equals the sum of the two remote interior angles, no need to find the third one. In an isosceles triangle the base angles are equal — half of all AMC angle chases run on this fact plus 180° .

Polygons. For regular polygons, compute through the EXTERIOR angle: it equals $360^\circ/n$, and the interior angle is $180^\circ - 360^\circ/n$. Given an interior angle of 156° ? Do not solve an equation with $(n - 2)$: the exterior angle is 24° , so $n = 360/24 = 15$. Faster, and no arithmetic potholes.

Bisectors. Angle bisectors appear in angle chases all the time: express their half-angles through the 180° sum instead of memorizing ready-made facts — one clean line of arithmetic beats a misremembered formula.

WORKED EXAMPLES

EXAMPLE 1 • PARALLEL LINES

Lines a and b are parallel. A transversal makes a 65° angle with line a . Find the same-side interior angle at line b .

→ Same-side interior angles add to 180° : the angle is $180^\circ - 65^\circ = 115^\circ$. The trap is mixing up the pairs: the alternate interior angle would be 65° . Before answering, take one second to check whether your pair gives EQUALITY or a SUPPLEMENT.

EXAMPLE 2 • EXTERIOR ANGLE

In triangle ABC , angle $A = 50^\circ$ and the exterior angle at vertex B is 120° . Find angle C .

→ The exterior angle at B equals the sum of the remote interior angles: $120^\circ = A + C = 50^\circ + C$, so $C = 70^\circ$. The slow route — first find $B = 60^\circ$, then subtract from 180° — gives the same answer in twice as many moves.

EXAMPLE 3 • REGULAR POLYGON

The interior angle of a regular polygon is 156° . How many sides does it have?

→ Exterior angle: $180^\circ - 156^\circ = 24^\circ$. All exterior angles sum to 360° , so $n = 360/24 = 15$. Check via $(n - 2) \cdot 180$: $13 \cdot 180 = 2340$ and $2340/15 = 156$. Correct. The exterior angle is always the shorter road.

EXAMPLE 4 • ANGLE CHASE

In triangle ABC , $AB = AC$ and angle $A = 36^\circ$. The bisector of angle B meets side AC at point D . Find angle BDC .

→ Base angles: $(180^\circ - 36^\circ)/2 = 72^\circ$. The bisector halves angle B : angle $ABD = 36^\circ$. In triangle ABD : angle $ADB = 180^\circ - 36^\circ - 36^\circ = 108^\circ$, so the supplementary angle $BDC = 72^\circ$. One method for every chase: write a number on EVERY angle as you find it; keep nothing in your head.

INDEPENDENT SET

1. The angles of a triangle are in the ratio $2 : 3 : 4$. Find the largest angle.
2. Lines a and b are parallel. The same-side interior angles at a transversal are $(3x + 20)^\circ$ and $(2x - 10)^\circ$. Find x .
3. The exterior angle of a regular polygon is 24° . How many sides does it have?
4. The apex angle of an isosceles triangle is 40° . Find a base angle.
5. The interior angles of a convex polygon sum to 1440° . How many sides does it have?
6. An exterior angle of a triangle is 110° , and one of the remote interior angles is 45° . Find the other remote interior angle.
7. In a regular polygon, each interior angle is 5 times the exterior angle. How many sides does the polygon have?
8. Point P lies between parallel lines a and b . A segment runs from point A on line a to P , and a segment runs from P to point B on line b . The angle between AP and line a is 35° , and the angle between PB and line b is 45° (both angles open toward P). Find angle APB .
9. ★ In triangle ABC , angle $A = 40^\circ$. The bisectors of angles B and C meet at point I . Find angle BIC .

Lesson 2.2. Similarity and Pythagoras: Special Triangles, Altitude to the Hypotenuse, Triples

THEORY

Similarity. Two equal angles — and the triangles are similar (the third angle comes free). All linear measurements of similar figures share one ratio k : sides, altitudes, perimeters. Areas scale by k^2 . A line parallel to a side of a triangle cuts off a similar triangle: $DE \parallel BC$ immediately gives $AD/AB = AE/AC = DE/BC$.

Pythagoras and triples. In a right triangle, $a^2 + b^2 = c^2$. Memorize the triples: 3-4-5, 5-12-13, 7-24-25, 8-15-17 — and their multiples (6-8-10, 9-12-15, 10-24-26). On the AMC, recognizing a triple is faster than computing the root; see a leg of 7 and a hypotenuse of 25 — the other leg is 24, no computation.

45-45-90: sides $x : x : x\sqrt{2}$. 30-60-90: sides $x : x\sqrt{3} : 2x$ (x is opposite 30° , and $x\sqrt{3}$ is opposite 60°).

Special triangles. Half a square is a 45-45-90: diagonal = side $\cdot \sqrt{2}$. Half an equilateral triangle is a 30-60-90. The classic mistake is attaching $\sqrt{3}$ to the wrong side: $\sqrt{3}$ always belongs to the MIDDLE side, the one opposite 60° .

Altitude to the hypotenuse. It cuts a right triangle into two triangles similar to the original. Hence three formulas: $h = ab/c$ (two expressions for the area), $h^2 = mn$, and $\text{leg}^2 = c \cdot (\text{its own projection})$, where m, n are the segments of the hypotenuse. These three identities close out an entire class of AMC problems.

WORKED EXAMPLES

EXAMPLE 1 • 30-60-90

A right triangle has an acute angle of 30° and a hypotenuse of 10. Find both legs.

→ The pattern is $x : x\sqrt{3} : 2x$. The hypotenuse $2x = 10$, so $x = 5$: the legs are **5 and $5\sqrt{3}$** . Pythagorean check: $25 + 75 = 100$. Correct. The trap is placing $5\sqrt{3}$ opposite the 30° angle: the smaller angle faces the smaller side.

EXAMPLE 2 • SIMILARITY

In triangle ABC , point D is on side AB and point E is on side AC , with $DE \parallel BC$. Given: $AD = 4$, $DB = 6$, $DE = 6$. Find BC .

→ Triangles ADE and ABC are similar with ratio $AD/AB = 4/10 = 2/5$. So $BC = DE \cdot 5/2 = \mathbf{15}$. The classic trap is taking $AD/DB = 4/6$: the similarity ratio compares to the WHOLE side, not to its piece.

EXAMPLE 3 • ALTITUDE TO THE HYPOTENUSE

The legs of a right triangle are 6 and 8. Find the altitude to the hypotenuse and the segments into which it divides the hypotenuse.

→ The 6-8-10 triple: the hypotenuse is 10. Altitude from the area: $h = 6 \cdot 8/10 = \mathbf{4.8}$. Projections of the legs: $6^2/10 = 3.6$ and $8^2/10 = 6.4$. Check: $3.6 + 6.4 = 10$ and $h^2 = 3.6 \cdot 6.4 = 23.04 = 4.8^2$. Everything agrees.

EXAMPLE 4 • TRIPLES

A right triangle has hypotenuse 25 and one leg equal to 7. Find the perimeter.

→ This is the 7-24-25 triple: the other leg is **24**, and the perimeter is $7 + 24 + 25 = \mathbf{56}$. Without the triple you would compute $\sqrt{(625 - 49)} = \sqrt{576}$ — doable, but a minute slower. Memorized triples are purchased time.

INDEPENDENT SET

1. The legs of a right triangle are 9 and 12. Find the hypotenuse.
2. The diagonal of a square is 10. Find its area.
3. In a 30-60-90 triangle the shorter leg is 7. Find the hypotenuse.
4. A triangle has sides 6, 8, 10. A similar triangle has perimeter 60. Find its longest side.
5. In triangle ABC , segment $DE \parallel BC$ with D on AB , $AD : DB = 2 : 3$, and $BC = 20$. Find DE .
6. The legs of a right triangle are 5 and 12. Find the altitude to the hypotenuse.
7. The diagonal of a rectangle is 17, and one side is 8. Find the area of the rectangle.
8. A 25-foot ladder leans against a wall with its foot 7 feet from the wall. The top slides down 4 feet. How far does the foot slide away from the wall?
9. ★ A right triangle has legs 15 and 20. The altitude from the right angle meets the hypotenuse at point D . Find the distance from D to the leg of length 15.

Lesson 2.3. Areas: Triangle, Trapezoid, Complete-and-Subtract, Ratios

THEORY

Triangle. The base formula is half of base times height: $S = \frac{1}{2}ah$. ANY side can serve as the base — pick the one whose altitude you can see immediately. For a right triangle, the area is half the product of the legs, no altitude hunting needed.

Heron: $S = \sqrt{p(p-a)(p-b)(p-c)}$, where p is the semiperimeter. Trapezoid: $S = \frac{a+b}{2} \cdot h$.

Heron and the usual suspects. The 13-14-15 triangle (area 84) and its relatives 5-5-6 and 9-10-17 have lived in contest problems for decades. Often Heron is overkill: the altitude of an isosceles triangle falls out of Pythagoras in one move. A rhombus, or any quadrilateral with perpendicular diagonals: $S =$ half the product of the diagonals.

Complete-and-subtract. Do not attack a crooked polygon head-on: complete it to a rectangle and subtract the extra right triangles at the corners. This method beats any formula on figures with lattice-point vertices, and it trains your hand for the Shoelace formula of Lesson 2.5.

Area ratios. Triangles with a shared altitude compare as their bases: a cevian AD with $BD : DC = m : n$ splits the area in the ratio $m : n$. A median halves the area; the midsegment triangle takes $\frac{1}{4}$ of it. Two successive ratios multiply — the main lever of AMC #10–15 area problems.

WORKED EXAMPLES

EXAMPLE 1 • HERON

Find the area of the triangle with sides 13, 14, 15.

→ Semiperimeter $p = 21$. $S = \sqrt{(21 \cdot 8 \cdot 7 \cdot 6)} = \sqrt{7056} = \mathbf{84}$. Take the root by grouping: $21 \cdot 7 = 147 = 49 \cdot 3$ and $8 \cdot 6 = 48 = 16 \cdot 3$, so $S = 7 \cdot 4 \cdot 3 = 84$. Bonus: the altitude to the side of 14 is $2 \cdot 84/14 = 12$.

EXAMPLE 2 • TRAPEZOID

A trapezoid has bases 8 and 14 and height 5. Find its area.

→ $S = (8 + 14)/2 \cdot 5 = 11 \cdot 5 = \mathbf{55}$. The half-sum 11 is the midsegment: the trapezoid has the same area as an 11-by-5 rectangle. The trap is dropping the division by 2 and getting 110: a decoy of exactly that shape is almost always among the choices.

EXAMPLE 3 • COMPLETE-AND-SUBTRACT

One corner is cut off a 12-by-9 rectangle: a right triangle with legs 4 and 3 lying along the sides of the rectangle. Find the area of the remaining pentagon.

→ The whole minus the extra: $12 \cdot 9 - 4 \cdot 3/2 = 108 - 6 = \mathbf{102}$. The same move runs in reverse: complete a nonconvex figure to a rectangle and subtract the added pieces. One big simple figure minus simple pieces — instead of one complicated figure.

EXAMPLE 4 • AREA RATIOS

Triangle ABC has area 60. Point D lies on side BC with $BD : DC = 2 : 3$; point E is the midpoint of segment AD . Find the area of triangle ABE .

→ Step 1: shared altitude from A , so $[ABD] = 60 \cdot 2/5 = 24$. Step 2: BE is a median of triangle ABD and halves its area: $[ABE] = \mathbf{12}$. Two ratios multiplied — and not a single altitude was computed.

INDEPENDENT SET

1. A triangle has base 10 and the altitude to it equal to 7. Find the area.
2. A trapezoid has bases 6 and 10 and height 4. Find its area.
3. Find the area of the triangle with sides 5, 5, 6.
4. Find the area of the triangle with sides 9, 10, 17.
5. The diagonals of a rhombus are 10 and 24. Find its area.
6. Triangle ABC has area 54. Point D lies on side BC with $BD : DC = 4 : 5$. Find the area of triangle ABD .
7. A triangle has area 40. Find the area of the triangle formed by the midpoints of its sides.
8. Triangle ABC has area 81. Point M is on side AB and point N is on side AC , with $AM : MB = 1 : 2$ and $AN : NC = 1 : 2$. Find the area of quadrilateral $BMNC$.
9. ★ In trapezoid $ABCD$, the bases are $BC = 5$ and $AD = 15$, and the diagonals meet at point O . The area of triangle BOC is 3. Find the area of the trapezoid.

Lesson 2.4. Circles: Inscribed Angles, Tangents, $r = S/p$

THEORY

Inscribed and central angles. An inscribed angle is half the central angle on the same arc; all inscribed angles on one arc are equal. The special case that settles hundreds of problems: an angle subtending a diameter is right (Thales). In a cyclic quadrilateral, opposite angles sum to 180° .

Tangents. A tangent is perpendicular to the radius at the point of tangency — see a tangent, drop a right angle and ready Pythagoras. Two tangent segments from one external point are equal; the tangent length from a point at distance d from the center is $\sqrt{d^2 - r^2}$.

Inscribed angle = $\frac{\text{central angle}}{2}$. Incircle: $r = S/p$ (p is the semiperimeter). Right triangle: $r = \frac{a+b-c}{2}$, $R = c/2$.

Incircle and circumcircle. Cut the triangle into three triangles with a common vertex at the incenter: each has height r , whence $S = pr$ and $r = S/p$. In a right triangle, the circumcenter is the midpoint of the hypotenuse: $R = c/2$, which instantly gives the median to the hypotenuse.

Chords. The perpendicular from the center bisects a chord: half the chord, the radius, and the distance to the center form a right triangle. A chord of 8 in a circle of radius 5 sits at distance 3 from the center — the same 3-4-5 triple in a new costume.

WORKED EXAMPLES

EXAMPLE 1 • INSCRIBED ANGLE

Chord AB subtends a central angle of 100° . Point C lies on the major arc AB . Find angle ACB .

→ Inscribed angle ACB stands on the same arc as the central angle: $100^\circ/2 = 50^\circ$. The trap is a point on the MINOR arc: there the angle would stand on the 260° arc and equal 130° . Always check which arc the vertex faces.

EXAMPLE 2 • TANGENT

Point P is at distance 13 from the center of a circle of radius 5. Find the length of the tangent from P .

→ The radius to the point of tangency is perpendicular to the tangent: a right triangle with hypotenuse 13 and leg 5. Tangent = $\sqrt{(169 - 25)} = 12$ — the 5-12-13 triple. Every tangent problem starts with this right angle.

EXAMPLE 3 • INCIRCLE

Find the radius of the circle inscribed in a right triangle with legs 6 and 8.

→ The hypotenuse is 10. Via $r = S/p$: area 24, semiperimeter 12, so $r = 24/12 = 2$. Quick check with the right-triangle formula: $(6 + 8 - 10)/2 = 2$. Agreed. Keep both formulas ready: the first is universal, the second instantaneous.

EXAMPLE 4 • CYCLIC QUADRILATERAL

Quadrilateral $ABCD$ is inscribed in a circle, with angle $A = 70^\circ$ and angle $B = 85^\circ$. Find angles C and D .

→ Opposite angles sum to 180° : $C = 180^\circ - 70^\circ = 110^\circ$, $D = 180^\circ - 85^\circ = 95^\circ$. Check: $70 + 85 + 110 + 95 = 360$. Correct. The pairs are the OPPOSITE angles: A with C , B with D , not adjacent ones.

INDEPENDENT SET

1. A central angle is 80° . Find the inscribed angle on the same arc.
2. Segment AB is a diameter of a circle, point C lies on the circle, and angle $CAB = 35^\circ$. Find angle CBA .
3. Point P is at distance 10 from the center of a circle of radius 6. Find the tangent length from P .
4. A quadrilateral is inscribed in a circle, and one of its angles is 100° . Find the opposite angle.
5. Find the radius of the circle inscribed in the triangle with sides 13, 14, 15.

6. The legs of a right triangle are 9 and 12. Find the circumradius.
7. Tangents PA and PB are drawn from point P to a circle with center O ; angle $APB = 40^\circ$. Find central angle AOB .
8. In a circle of radius 5, two parallel chords of lengths 6 and 8 lie on the same side of the center. Find the distance between them.
9. ★ The circle inscribed in a right triangle touches the hypotenuse at a point dividing it into segments of lengths 6 and 4. Find the area of the triangle.

Lesson 2.5. Coordinate Geometry: Distance, Lines, Circles, Shoelace

THEORY

Distance and midpoint. The distance between points is Pythagoras on coordinate differences: $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$. The midpoint takes half-sums of the coordinates. Differences of 6 and 8, 5 and 12, 8 and 15 — recognize the triples instead of grinding out the square root.

Lines. Slope $k = \Delta y / \Delta x$: the rise per step right. Parallel lines: equal slopes; perpendicular: $k_1 k_2 = -1$. A line from a point and a slope: $y - y_0 = k(x - x_0)$. Axis intercepts come from substituting zero — by substitution, not from memory.

Shoelace for a triangle $(x_1, y_1), (x_2, y_2), (x_3, y_3)$: $S = \frac{1}{2} |x_1 y_2 - x_2 y_1 + x_2 y_3 - x_3 y_2 + x_3 y_1 - x_1 y_3|$. Circle: $(x - a)^2 + (y - b)^2 = r^2$.

Circles. An equation with $x^2 + y^2$ and linear terms is tamed by completing the square: from $x^2 - 6x$ we get $(x - 3)^2 - 9$. The center and radius read off the standard form; do not forget to carry the leftover constants to the right-hand side.

Shoelace. Walk the vertices strictly along the boundary (clockwise or counterclockwise, but never jumping across the figure); the absolute value at the end fixes the sign. Put a vertex with zeros first: zero terms burn away and the computation halves. For any polygon the formula is the same: consecutive vertex pairs around the loop.

WORKED EXAMPLES

EXAMPLE 1 • DISTANCE AND MIDPOINT

Given points $A(-2, 3)$ and $B(4, 11)$, find the length of segment AB and its midpoint.

→ Differences: $\Delta x = 6$, $\Delta y = 8$ — the 6-8-10 triple, so the length is **10** with no root. Midpoint: $((-2 + 4)/2, (3 + 11)/2) =$ **(1, 7)**. The common midpoint error is a half-difference instead of a half-sum: the midpoint is the AVERAGE of the coordinates.

EXAMPLE 2 • EQUATION OF A LINE

A line passes through $(2, 3)$ and $(6, 11)$. Find its equation and the slope of a line perpendicular to it.

→ Slope: $(11 - 3)/(6 - 2) = 2$. Through $(2, 3)$: $y - 3 = 2(x - 2)$, that is **$y = 2x - 1$** . Check with the second point: $2 \cdot 6 - 1 = 11$. Correct. Perpendicular slope: $k = -1/2$ (the product of slopes is -1).

EXAMPLE 3 • SHOELACE

Find the area of the triangle with vertices $(0, 0)$, $(8, 2)$, $(3, 7)$.

→ With the zero vertex first, a single cross product survives: $S = |8 \cdot 7 - 3 \cdot 2|/2 = |56 - 6|/2 =$ **25**. The full Shoelace gives the same: $0 \cdot 2 - 8 \cdot 0 + 8 \cdot 7 - 3 \cdot 2 + 3 \cdot 0 - 0 \cdot 7 = 50$, half of which is 25.

EXAMPLE 4 • CIRCLE

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

→ Complete the squares: $(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12$, so $(x - 3)^2 + (y + 2)^2 = 25$. Center **(3, -2)**, radius **5**. Two snares: the center's sign is opposite to the sign inside the parentheses, and 25 is r^2 , not the radius.

INDEPENDENT SET

1. Find the distance between (1, 2) and (7, 10).
2. Find the midpoint of the segment with endpoints (-3, 5) and (9, -1).
3. Find the slope of the line through (2, -1) and (5, 8).
4. The line $3x + 4y = 12$ cuts a triangle off the coordinate axes. Find its area.
5. A line is perpendicular to $y = 2x + 3$ and passes through (4, 1). Find its y -intercept.
6. Find the radius of the circle $x^2 + y^2 - 4x + 6y - 3 = 0$.
7. Find the area of the quadrilateral with vertices (0, 0), (5, 0), (6, 4), (1, 3), listed along the boundary.
8. A point on the x -axis is equidistant from $A(1, 2)$ and $B(7, 4)$. Find its x -coordinate.
9. ★ Points $A(2, 3)$ and $B(10, 9)$ lie on the same side of the x -axis. Point P is chosen on the x -axis so that $AP + PB$ is as small as possible. Find the x -coordinate of P .

Lesson 2.6. Minimal Solid Geometry: Volumes, Surfaces, Nets, the Box Diagonal

THEORY

Prism and cylinder. Anything “extruded” from a base along a height: volume = base area $\times$ height. Cylinder: $V = \pi r^2 h$, the lateral surface unrolls into a $2\pi r$ -by- h rectangle, and the total surface is $2\pi r h + 2\pi r^2$. A cube of edge a : volume a^3 , surface $6a^2$.

Cone, pyramid, sphere. Everything “pointed” takes a third of its prism: $V = \frac{1}{3} \cdot \text{base} \cdot \text{height}$. In a cone, the radius, height, and slant l are tied by Pythagoras: $l^2 = r^2 + h^2$; the lateral surface is $\pi r l$. Sphere: $V = \frac{4}{3}\pi r^3$, surface $4\pi r^2$.

Diagonal of an $a \times b \times c$ box: $d = \sqrt{a^2 + b^2 + c^2}$. Cube: $d = a\sqrt{3}$.

Nets. A cylinder's lateral surface unrolls into a rectangle with sides $2\pi r$ and h ; a cone's lateral surface unrolls into a sector of radius l whose arc length is $2\pi r$. The shortest path along a surface is a straight segment on the net — it looks curved in 3D, but it is straight once unrolled.

The diagonal and water levels. The box diagonal is a double Pythagoras in one formula. Submersion and pouring problems rest on one law: volume does not vanish. A body fully submerged in a cylinder raises the level by exactly its volume divided by the base area.

WORKED EXAMPLES

EXAMPLE 1 • BOX DIAGONAL

Find the diagonal of a 3-by-4-by-12 rectangular box.

→ $d = \sqrt{9 + 16 + 144} = \sqrt{169} = \mathbf{13}$. Two triples hide inside: 3-4-5 across the bottom and 5-12-13 in the vertical cross-section — the formula just glues two Pythagorean steps into one.

EXAMPLE 2 • CYLINDER

A cylinder has radius 3 and height 5. Find its volume and total surface area.

→ Volume: $\pi \cdot 9 \cdot 5 = \mathbf{45\pi}$. Surface: the side $2\pi \cdot 3 \cdot 5 = 30\pi$ plus two lids $2 \cdot 9\pi = 18\pi$, total $\mathbf{48\pi}$. AMC answers usually stay in terms of π — do not rush to multiply by 3.14.

EXAMPLE 3 • CONE

A cone has base radius 6 and height 8. Find its volume and lateral surface area.

→ Volume: $(1/3)\pi \cdot 36 \cdot 8 = \mathbf{96\pi}$. Slant: $\sqrt{36 + 64} = 10$ (the 6-8-10 triple), so the lateral surface is $\pi rl = \mathbf{60\pi}$. The trap is plugging the height into πrl instead of the slant: the surface formula runs on l .

EXAMPLE 4 • SPHERE

A sphere has radius 3. Find its volume and surface area.

→ Volume: $(4/3)\pi \cdot 27 = \mathbf{36\pi}$. Surface: $4\pi \cdot 9 = \mathbf{36\pi}$. The numbers match — a quirk of radius 3, not a law of nature. Keep the formulas separate: $(4/3)\pi r^3$ and $4\pi r^2$ get confused more than any other pair.

INDEPENDENT SET

1. A cube has edge 4. Find its surface area.
2. Find the diagonal of a 2-by-3-by-6 rectangular box.
3. A right prism has a right-triangle base with legs 3 and 4, and the prism's height is 10. Find the total surface area.
4. A sphere has surface area 100π . Find its volume.
5. A cone has base radius 5 and slant height 13. Find its volume.
6. A cube has volume 27. Find its space diagonal (vertex to opposite vertex).
7. The lateral surface of a cylinder with radius 4 and height 10 is unrolled into a rectangle. Find its area.
8. A sphere of radius 3 is fully submerged in a cylindrical tank of base radius 6 (the water covers it completely and does not overflow). By how much does the water level rise?
9. ★ A 216° sector of a circle of radius 10 is rolled into the lateral surface of a cone. Find the volume of the cone.

Solutions to the independent sets

LESSON 2.1

1. One part = $180/9 = 20^\circ$: the angles are 40° , 60° , 80° . The largest is **80°** .
 2. $5x + 10 = 180$, so $x = \mathbf{34}$ (the angles are 122° and 58°).
 3. $360/24 = \mathbf{15}$ sides.
 4. $(180 - 40)/2 = \mathbf{70^\circ}$.
 5. $n - 2 = 8$, so $n = \mathbf{10}$.
 6. $110 - 45 = \mathbf{65^\circ}$.
 7. 6 parts = 180° , so the exterior angle is 30° : $n = 360/30 = \mathbf{12}$.
 8. The auxiliary parallel through P splits angle APB into two alternate interior angles: $35^\circ + 45^\circ = \mathbf{80^\circ}$.
- ★ Angles $B + C = 140^\circ$, and their halves give 70° , so angle $BIC = 180^\circ - 70^\circ = \mathbf{110^\circ}$. The general fact: the angle between the bisectors is $90^\circ + A/2$.

LESSON 2.2

1. 3-4-5 scaled by 3: the hypotenuse is **15**.
 2. Area = $d^2/2 = 100/2 = \mathbf{50}$.
 3. $2 \cdot 7 = \mathbf{14}$ (and the longer leg is $7\sqrt{3}$).
 4. Ratio $60/24 = 2.5$: the sides are 15, 20, 25. The longest is **25**.
 5. $AD/AB = 2/5$, so $DE = 20 \cdot 2/5 = \mathbf{8}$.
 6. The hypotenuse is 13 (the 5-12-13 triple): $h = 5 \cdot 12/13 = \mathbf{60/13}$.
 7. The 8-15-17 triple: the other side is 15, so the area is $8 \cdot 15 = \mathbf{120}$.
 8. First the height: $\sqrt{625 - 49} = 24$ (the 7-24-25 triple). New height 20, new base $\sqrt{625 - 400} = 15$ (the 15-20-25 triple). The slide: $15 - 7 = \mathbf{8}$.
- ★ The hypotenuse is 25. The segment next to the leg 15: $AD = 15^2/25 = 9$. The perpendicular from D to that leg creates a triangle similar to the original: distance = $AD \cdot 20/25 = 9 \cdot 0.8 = \mathbf{7.2}$. Two similarity steps in a row is the signature move of problems #11–15.

LESSON 2.3

1. $10 \cdot 7/2 = \mathbf{35}$.
 2. $(6 + 10)/2 \cdot 4 = \mathbf{32}$.
 3. Altitude $\sqrt{25 - 9} = 4$: area $6 \cdot 4/2 = \mathbf{12}$.
 4. $\sqrt{(18 \cdot 9 \cdot 8 \cdot 1)} = \sqrt{1296} = \mathbf{36}$.
 5. $10 \cdot 24/2 = \mathbf{120}$ (and the side of the rhombus is 13, the 5-12-13 triple).
 6. $54 \cdot 4/9 = \mathbf{24}$.
 7. The area ratio is $(1/2)^2 = 1/4$: area **10**.
 8. $[AMN] = 81 \cdot (1/3)(1/3) = 9$, so $[BMNC] = 81 - 9 = \mathbf{72}$. Subtracting beats computing the quadrilateral directly.
- ★ The 1 : 3 similarity gives $[DOA] = 9 \cdot 3 = 27$. Then $OA : OC = 3 : 1$, so $[AOB] = 3 \cdot [BOC] = 9$, and likewise $[COD] = 9$. Total: $3 + 27 + 9 + 9 = \mathbf{48}$.

LESSON 2.4

1. $80/2 = 40^\circ$.
 2. Angle $C = 90^\circ$ (Thales), so $CBA = 90 - 35 = 55^\circ$.
 3. $\sqrt{100 - 36} = 8$ (the 6-8-10 triple).
 4. $180 - 100 = 80^\circ$.
 5. $r = S/p = 84/21 = 4$.
 6. The hypotenuse is 15 (the 9-12-15 triple), so $R = 7.5$.
 7. $360 - 90 - 90 - 40 = 140^\circ$.
 8. The chord of 6 sits at distance $\sqrt{25 - 9} = 4$, the chord of 8 at $\sqrt{25 - 16} = 3$. Same side: $4 - 3 = 1$. The trap is adding to get 7: that answers the OPPOSITE-sides version.
- ★ Pythagoras: $(6 + r)^2 + (4 + r)^2 = 100$, that is $r^2 + 10r - 24 = 0$, so $r = 2$. The legs are 8 and 6: area $8 \cdot 6/2 = 24$. Check: the 6-8-10 triple.

LESSON 2.5

1. The 6-8-10 triple: distance **10**.
 2. $((-3 + 9)/2, (5 - 1)/2) = (3, 2)$.
 3. $(8 - (-1))/(5 - 2) = 9/3 = 3$.
 4. Intercepts (4, 0) and (0, 3): a right triangle of area $4 \cdot 3/2 = 6$.
 5. $y - 1 = -(x - 4)/2$, that is $y = -x/2 + 3$: the intercept is **(0, 3)**.
 6. $(x - 2)^2 + (y + 3)^2 = 3 + 4 + 9 = 16$: radius **4**.
 7. The cross-product sum: $0 + (5 \cdot 4 - 6 \cdot 0) + (6 \cdot 3 - 1 \cdot 4) + 0 = 20 + 14 = 34$. Area $34/2 = 17$.
 8. $(x - 1)^2 + 4 = (x - 7)^2 + 16$: $-2x + 5 = -14x + 65$, so $12x = 60$ and $x = 5$. Check: both distances are $\sqrt{20}$.
- ★ Reflect: $A'(2, -3)$. Line $A'B$ has slope $12/8 = 3/2$ and crosses the x -axis where $-3 + (3/2)(x - 2) = 0$, i.e. $x = 4$. Every other path is longer than the reflection-straightened one — the signature shortest-path trick.

LESSON 2.6

1. $6 \cdot 16 = 96$.
 2. $\sqrt{4 + 9 + 36} = \sqrt{49} = 7$.
 3. The base is a 3-4-5 triangle: area 6, perimeter 12. Total surface: $2 \cdot 6 + 12 \cdot 10 = 132$.
 4. $r = 5$: volume $(4/3)\pi \cdot 125 = 500\pi/3$.
 5. $h = 12$: volume $(1/3)\pi \cdot 25 \cdot 12 = 100\pi$.
 6. Edge 3: diagonal $3\sqrt{3}$.
 7. $2\pi \cdot 4 \cdot 10 = 80\pi$.
 8. Sphere volume 36π , base area 36π : the rise is $36\pi/36\pi = 1$.
- ★ Arc: $(216/360) \cdot 2\pi \cdot 10 = 12\pi$, so the base radius is 6. The slant is 10, and the height is $\sqrt{100 - 36} = 8$. Volume: $(1/3)\pi \cdot 36 \cdot 8 = 96\pi$.